# Human Resources (HR Analytics)

*Domain Notes for Data Analysts / Data Scientists*

# 01. Overview — What is this domain?

HR Analytics (also called People Analytics or Workforce Analytics) covers the analytical work behind managing an organization's most expensive and most variable asset: its people. It applies data analysis and statistical/ML methods to workforce decisions that were traditionally made on intuition or policy alone. This domain includes:

- Talent Acquisition & Recruiting — sourcing, screening, and hiring candidates efficiently and fairly
- Employee Retention & Attrition — understanding and predicting why employees leave
- Performance Management — measuring and improving employee performance and productivity
- Compensation & Benefits Analytics — pay equity, market benchmarking, total rewards optimization
- Learning & Development (L&D) — training effectiveness and skills gap analysis
- Diversity, Equity & Inclusion (DEI) Analytics — representation, pay equity, and inclusion measurement
- Workforce Planning — headcount forecasting, succession planning, organizational design
- Employee Engagement & Experience — survey analytics, sentiment, and culture measurement

## Why data analytics/science matters here

**1.** People costs are usually the largest line item — even small improvements in retention or hiring efficiency have outsized financial impact.

**2.** Attrition is expensive and often preventable — replacing an employee can cost 50-200% of their annual salary when recruiting, onboarding, and lost productivity are counted.

**3.** Hiring at scale requires consistency — data-driven screening helps reduce bias and improve quality-of-hire compared to purely subjective judgment.

**4.** Engagement and culture directly affect productivity — quantifying "soft" concepts like engagement turns them into actionable, trackable metrics.

**5.** Regulatory and reputational risk around fairness — pay equity and hiring discrimination are legally and ethically sensitive areas requiring rigorous, defensible analysis.

## Industry context & key regulatory/ethical considerations

| Region | Relevant Regulation/Body |
|---|---|
| USA | EEOC (Equal Employment Opportunity Commission), FLSA (Fair Labor Standards Act), ADA |
| India | Equal Remuneration Act, Industrial Relations Code, POSH Act |
| UK/EU | GDPR (employee data privacy), Equality Act (UK), EU Pay Transparency Directive |
| Global | ILO (International Labour Organization) standards, ISO 30414 (Human Capital Reporting standard) |

Key concepts a HR analyst should be aware of:

- Adverse Impact Analysis — statistically testing whether a hiring/promotion process disproportionately screens out a protected group (commonly using the "four-fifths rule")
- Pay Equity Analysis — statistical testing (often regression-based) to detect unexplained pay gaps by gender, race, or other protected characteristics
- Employee Data Privacy — HR data is highly sensitive; access must be tightly controlled and anonymized/aggregated wherever possible

- ISO 30414 — an international standard for human capital reporting, increasingly referenced by large organizations for external workforce disclosures

### Typical business functions a Data Analyst/Scientist supports

- Talent Acquisition / Recruiting Analytics
- Retention & Attrition Modeling
- Compensation & Pay Equity Analysis
- Workforce Planning & Headcount Forecasting
- Employee Engagement & Survey Analytics
- Learning & Development Effectiveness Measurement
- DEI (Diversity, Equity & Inclusion) Reporting
- HR Operations & Process Analytics (time-to-hire, HR service desk metrics)

### Why this domain is attractive for Data Analysts/Scientists

- Highly visible, business-critical impact — attrition and hiring outcomes are watched closely by executive leadership
- A growing, maturing field — many organizations are still building out their first People Analytics teams, offering room to shape practice
- Rich blend of quantitative rigor and human/organizational nuance — requires translating statistics into decisions that affect real people fairly
- Career path: HR/People Analyst → Senior People Analyst → People Analytics/HR Data Scientist → Head of People Analytics/Chief People Officer (analytics-track)

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# 02. Role of a Data Analyst/Scientist — Day-to-Day & Full Project Lifecycle

## A. Day-to-Day Responsibilities (Snapshot)

- Extracting and validating employee data from the HRIS (Human Resource Information System)
- Building dashboards for headcount, attrition, and hiring pipeline tracking
- Analyzing exit interview data and survey results for patterns
- Building attrition/turnover prediction models
- Supporting compensation review cycles with pay equity analysis
- Preparing board/leadership-level workforce reporting (e.g., quarterly people metrics reviews)
- Partnering with HR Business Partners (HRBPs) to translate data findings into action plans
- Ensuring strict data privacy and access controls given the sensitivity of HR data

## B. End-to-End Project Lifecycle (Example: Employee Attrition Prediction Model)

### Step 1: Business Problem Understanding

Meet with HR/People leadership to understand the pain point.

*Example: "Our voluntary attrition in the engineering department has risen to 18% annually, well above our target of 10%, and we're losing high performers. We need to understand why and intervene before they leave."*

Translate into an analytical framing:

- Business ask → Reduce voluntary attrition, especially among high performers
- Analytical framing → Build a model to predict which current employees are at high risk of voluntarily leaving in the next 3-6 months, so managers/HR can intervene proactively

Define success metrics with stakeholders: model's ability to correctly flag at-risk employees (recall), actual reduction in attrition rate among flagged/intervened employees, and manager adoption of the tool.

## Step 2: Data Collection & Understanding

- Identify data sources: HRIS system (Workday, SAP SuccessFactors), performance management system, compensation data, engagement survey results, manager change history, timesheet/overtime data
- Understand what data is legitimately available and appropriate to use — HR data carries strong ethical obligations, and certain variables (e.g., protected characteristics) must be handled with extreme care to avoid discriminatory outcomes
- Define the target variable clearly: Attrition = 1 if employee voluntarily resigned within the prediction window; excluding involuntary terminations and retirements, which have very different drivers
- Clarify the prediction window (e.g., predicting risk of leaving in the next 6 months) and refresh cadence (e.g., monthly re-scoring)

## Step 3: Data Extraction

- Pull employee master data (tenure, role, level, department, location, manager)
- Pull compensation data (current salary, last raise %, salary relative to market/band)
- Pull performance data (recent ratings, promotion history, goal achievement)
- Pull engagement/survey data (eNPS scores, sentiment from open-text comments if available)
- Pull behavioral proxies where ethically and legally appropriate (e.g., internal job application activity, which can signal a search for other opportunities within the company)

## Step 4: Data Cleaning & Preprocessing

- Handle missing survey responses (non-response itself can be a signal, but must be handled carefully, not simply imputed as neutral)
- Standardize job titles/levels across different systems or historical naming conventions
- Handle organizational changes (mergers, restructuring) that may distort tenure or manager-change variables
- Address class imbalance — voluntary attrition is typically a minority class (often 10-20% annually)
- Exclude or separately model involuntary terminations, since mixing them with voluntary attrition would corrupt the target variable's meaning

## Step 5: Exploratory Data Analysis (EDA)

- Analyze attrition rate by tenure band (attrition often spikes at specific tenure milestones — e.g., around 1 year and again around 3-4 years)
- Analyze attrition by manager, department, compensation competitiveness, and performance rating
- Identify "regretted attrition" patterns — are high performers leaving at higher rates than low performers?
- Visualize time-to-attrition trends and seasonality (e.g., post-bonus-payout resignation spikes)

## Step 6: Feature Engineering

- Tenure-based features: months since hire, months since last promotion, months since last raise
- Compensation features: salary relative to internal peer band, salary relative to external market benchmark (compa-ratio), time since last salary increase

- Manager/team features: manager tenure, team size, recent manager change, team attrition rate (peer attrition can be contagious)
- Engagement features: latest eNPS/survey score, sentiment trend over time, participation rate in surveys
- Performance features: recent performance rating, rating trend (improving/declining), goal achievement history

## Step 7: Model Building

- Split data using a time-based approach — train on historical periods, validate on a more recent holdout period to simulate real deployment
- Common algorithms: Logistic Regression (interpretable baseline, easier to explain to HR stakeholders), Random Forest/XGBoost/LightGBM (for improved accuracy), Survival Analysis (Cox Proportional Hazards) for modeling time-to-attrition rather than a simple binary outcome
- Prioritize interpretability — HR and management stakeholders need to understand and trust why someone is flagged, not just receive a black-box score

## Step 8: Model Validation

- Evaluate using Precision/Recall (Recall matters greatly — missing a flight-risk high performer is costly) and AUC-ROC
- Validate on a later time period (temporal validation) to simulate real-world forward performance
- Explicitly test the model for fairness — ensure it does not disproportionately flag or overlook employees based on gender, age, race, or other protected characteristics; this is both an ethical and legal necessity
- Review flagged employees with HRBPs for face validity — do the top risk factors align with what HR partners are hearing on the ground?

## Step 9: Model Governance & Ethical Review

- Document model methodology, features used, and limitations clearly
- Exclude or carefully justify the use of any sensitive/protected variables
- Get sign-off from HR leadership, Legal, and/or an internal ethics/data governance committee before deployment, given the sensitivity of using predictive models on employees
- Establish clear guardrails on how the output will and will not be used (e.g., never used as sole justification for a negative employment action)

## Step 10: Deployment

- Integrate risk scores into a manager/HRBP-facing dashboard, not raw employee-level access — output should support conversations and retention actions, not surveillance
- Design a clear action framework: what should a manager/HRBP actually do when an employee is flagged (e.g., stay interview, career conversation, compensation review)
- Pilot with a single department/business unit before company-wide rollout to test workflow, response, and any unintended consequences

## Step 11: Monitoring & Maintenance

- Track model performance over time (precision/recall, calibration) and retrain periodically as workforce dynamics evolve
- Monitor real-world outcomes — are flagged/intervened employees actually staying at higher rates than a comparable non-intervened group?
- Watch for feedback loops or gaming (e.g., managers learning to game inputs) and adjust accordingly

## Step 12: Reporting & Stakeholder Communication

- Present attrition trends, root causes, and model impact to HR and business leadership in clear, non-technical language
- Provide actionable, aggregated insights back to the business (e.g., "compensation competitiveness is the top driver of attrition in the engineering org") rather than only individual-level scores
- Support board-level or ISO 30414-aligned human capital reporting where applicable

### C. This Lifecycle Applies (with Variations) to Other HR Use Cases

| Use Case | Key Lifecycle Differences |
|---|---|
| Recruiting/Candidate Screening Models | Target variable shifts to hiring success/quality-of-hire; extremely high scrutiny for adverse impact and legal compliance given hiring discrimination risk |
| Pay Equity Analysis | Less about prediction, more about regression-based statistical testing to detect unexplained pay gaps by protected group, usually run as a periodic compliance exercise rather than a continuously deployed model |
| Workforce Planning/Headcount Forecasting | Works at aggregate organizational level rather than individual employee level; combines demand forecasting (business growth) with supply forecasting (expected attrition) |
| Engagement Survey Analytics | Heavier emphasis on text/sentiment analysis of open-ended survey comments alongside quantitative scoring |

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## 03. Key Metrics & KPIs

### Talent Acquisition Metrics

| Metric | Meaning |
|---|---|
| Time-to-Fill | Average number of days from job requisition opening to offer acceptance |
| Time-to-Hire | Average number of days from candidate application to offer acceptance |
| Cost-per-Hire | Total recruiting cost divided by number of hires |
| Offer Acceptance Rate | % of extended offers that are accepted |
| Quality of Hire | Composite measure of new hire performance, retention, and manager satisfaction |
| Source of Hire Effectiveness | Which recruiting channels (referrals, job boards, agencies) produce the best hires |
| Applicant-to-Hire Ratio | Number of applicants required per successful hire, a funnel efficiency measure |

### Retention & Attrition Metrics

| Metric | Meaning |
|---|---|
| Voluntary Attrition Rate | % of employees who resign of their own accord, over a period |
| Involuntary Attrition Rate | % of employees terminated by the company |
| Regretted Attrition Rate | % of departures the company considers a loss (typically high performers) |
| First-Year Attrition Rate | % of new hires leaving within their first 12 months — signals hiring/onboarding issues |
| Retention Rate | % of employees retained over a given period (inverse of attrition) |
| Internal Mobility Rate | % of open roles filled by internal candidates |

### Engagement & Culture Metrics

| Metric | Meaning |
|---|---|
| Employee Net Promoter Score (eNPS) | Likelihood employees would recommend the company as a place to work |

| | |
|---|---|
| Engagement Survey Score | Composite score from structured employee engagement surveys |
| Survey Participation Rate | % of employees completing engagement/pulse surveys |
| Absenteeism Rate | % of scheduled work time lost to unplanned absence |

### Performance & Productivity Metrics

| Metric | Meaning |
|---|---|
| Performance Rating Distribution | Spread of performance ratings across the workforce (checking for rating inflation/bias) |
| Goal Achievement Rate | % of employees meeting or exceeding set performance goals |
| Revenue/Profit per Employee | Overall workforce productivity measure |
| Promotion Rate | % of employees promoted within a given period |

### Compensation & DEI Metrics

| Metric | Meaning |
|---|---|
| Compa-Ratio | Employee's salary relative to the midpoint of their salary band |
| Pay Equity Gap | Unexplained pay difference between protected and non-protected groups after controlling for legitimate factors |
| Salary Competitiveness Ratio | Company pay levels relative to external market benchmarks |
| Representation Metrics | % representation of gender/race/other groups across levels and functions |
| Promotion Parity | Whether promotion rates are equitable across demographic groups |

### Workforce Planning Metrics

| Metric | Meaning |
|---|---|
| Headcount Growth Rate | Rate of change in total employee count over time |
| Span of Control | Average number of direct reports per manager |
| Vacancy Rate | % of budgeted roles currently unfilled |
| Bench Strength/Succession Coverage | % of key roles with an identified ready-now successor |

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## 04. Common Datasets & Data Sources

### Internal Data Sources

| Source System | Typical Data |
|---|---|
| HRIS (Workday, SAP SuccessFactors, Oracle HCM) | Employee master data: role, level, department, tenure, manager, location |
| Applicant Tracking System (ATS) | Recruiting pipeline data: applications, interview stages, offers, sources |
| Performance Management System | Performance ratings, goals, 360-degree feedback |
| Compensation Management System | Salary, bonus, equity, pay bands, raise history |
| Learning Management System (LMS) | Training completion, course engagement, certifications |
| Engagement Survey Platforms | eNPS scores, survey responses, open-text comments |
| Time & Attendance Systems | Hours worked, overtime, absenteeism |

### External Data Sources

| Source | Typical Data |
|---|---|
| Compensation Benchmarking Providers (Mercer, Radford, Payscale) | External market salary data for competitive benchmarking |
| LinkedIn Talent Insights / LinkedIn Recruiter | Talent market supply/demand data, competitor hiring trends |
| Glassdoor/Indeed | Employer reviews, employee sentiment about competitors and own company |
| Government Labor Statistics (BLS, Ministry of Labour) | Broader labor market trends, unemployment rates |

### Typical Fields/Attributes an Analyst Works With

- Employee Data: employee ID, hire date, termination date and reason, job title, level, department, location, manager ID, tenure
- Compensation Data: base salary, bonus target/actual, equity grants, compa-ratio, last increase date/amount
- Performance Data: rating history, goal completion, promotion/demotion history
- Recruiting Data: requisition ID, source, application date, interview stage outcomes, offer date, offer acceptance
- Survey Data: eNPS score, engagement dimension scores (e.g., manager relationship, career growth), free-text comments

### Data Characteristics Specific to HR Analytics

- Extremely sensitive/PII data — compensation, performance, and demographic data require the strictest access controls
- Small numbers at granular levels — analyzing a specific team or demographic subgroup often means very small sample sizes, requiring careful statistical caution
- High risk of legal/ethical exposure — improper use of protected characteristics (even unintentionally, via proxy variables) can create discrimination liability
- Survey data has response bias — engagement scores reflect only those who chose to respond, which is rarely a random sample
- Organizational change disrupts historical data — restructures, M&A, and system migrations frequently break historical continuity in HR datasets

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## 05. Tools & Technologies

### Programming Languages

- Python — Pandas, Scikit-learn, Statsmodels, lifelines (for survival/attrition analysis)
- R — widely used for statistical analysis, especially pay equity regression analysis
- SQL — for querying HRIS/data warehouse tables

### HR-Specific Platforms

- Workday, SAP SuccessFactors, Oracle HCM Cloud — core HRIS platforms and source systems
- Visier — a leading dedicated people analytics platform with built-in benchmarking and predictive models
- Culture Amp, Glint (Microsoft Viva Glint), Qualtrics EX — engagement survey and analytics platforms
- Greenhouse, Lever, iCIMS — Applicant Tracking Systems (ATS) with built-in recruiting analytics

### Visualization & BI Tools

- Tableau, Power BI — for people analytics dashboards and executive reporting
- Excel — still heavily used for compensation modeling and ad-hoc HR reporting

### Statistical & ML Libraries

- Scikit-learn, XGBoost — for attrition/performance prediction models
- Statsmodels — for regression-based pay equity analysis
- lifelines (Python) / survival package (R) — for survival analysis of time-to-attrition
- spaCy/NLTK or transformer-based sentiment models — for analyzing open-text survey comments

### Data Infrastructure

- Snowflake, BigQuery, Redshift — for consolidating HR data from multiple systems into a unified warehouse
- dbt, Airflow — for data pipeline transformation and orchestration
- Strict role-based access control (RBAC) systems — essential given the sensitivity of HR data

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## 06. Real-World Use Cases

### Talent Acquisition

- Candidate Screening/Resume Scoring Models — predicting candidate-role fit, with heavy fairness/adverse-impact scrutiny
- Sourcing Channel Effectiveness Analysis — identifying which recruiting channels yield the best quality-of-hire
- Time-to-Fill Forecasting — predicting hiring timelines to support workforce planning

### Retention & Engagement

- Attrition Prediction — as detailed in the lifecycle example above
- Engagement Survey Driver Analysis — identifying which factors (manager relationship, career growth, compensation) most influence engagement scores
- Exit Interview Text Analysis — using NLP to identify common themes in why employees leave

### Compensation & DEI

- Pay Equity Analysis — regression-based testing for unexplained pay gaps across gender/race/other groups
- Compensation Benchmarking — comparing internal pay levels against external market data to stay competitive
- Promotion Parity Analysis — checking whether promotion rates are equitable across demographic groups

### Workforce Planning

- Headcount Forecasting — projecting future staffing needs based on business growth plans and expected attrition
- Succession Planning Analytics — identifying readiness and coverage for key leadership roles
- Organizational Network Analysis — mapping informal collaboration patterns to identify key connectors and silos

### Performance & Learning

- Performance Rating Calibration Analysis — checking for rating inflation or bias across managers/departments
- Training Effectiveness Measurement — evaluating whether L&D programs improve performance or retention
- Skills Gap Analysis — identifying mismatches between current workforce skills and future business needs

### Example Interview-Style Problem Statements

- "Build a model to predict which employees are at high risk of voluntarily leaving in the next 6 months."
- "How would you design a pay equity analysis to check for unexplained gender pay gaps in our organization?"
- "Our engineering team's engagement scores dropped 15 points this quarter — how would you investigate the cause?"
- "Design a dashboard to help HR business partners track hiring pipeline health across departments."

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## 07. ML & Statistical Techniques

### Classification & Prediction Models

- Logistic Regression — the standard interpretable baseline for attrition and hiring outcome prediction, often preferred by HR stakeholders
- Random Forest / Gradient Boosting (XGBoost, LightGBM) — for improved predictive accuracy on attrition and quality-of-hire models
- Survival Analysis (Kaplan-Meier, Cox Proportional Hazards) — modeling time-to-attrition rather than a simple binary outcome, capturing when employees are most likely to leave

### Pay Equity & Regression Techniques

- Multiple Linear Regression — the core method for pay equity analysis, regressing salary on legitimate factors (role, level, tenure, performance, location) and testing whether protected-class variables remain statistically significant after controlling for these factors
- Multilevel/Hierarchical Regression — accounting for nested organizational structure (employees within teams within departments) in compensation analysis

### Text & Sentiment Analysis

- Sentiment Analysis — extracting positive/negative sentiment from open-text survey comments and exit interviews
- Topic Modeling (LDA, BERTopic) — identifying common themes across large volumes of qualitative feedback
- Named Entity Recognition — extracting mentions of specific issues (e.g., manager names, specific policies) from free-text feedback for aggregate theme analysis

### Clustering & Segmentation

- K-Means/Hierarchical Clustering — segmenting employees into personas based on engagement, tenure, and performance patterns for targeted retention strategies
- RFM-style Segmentation adapted for HR — e.g., segmenting by recency of promotion, frequency of manager change, and compensation competitiveness

### Forecasting

- Time-Series Forecasting (ARIMA, Prophet) — for headcount and hiring demand forecasting
- Monte Carlo Simulation — for succession planning and workforce scenario modeling under uncertainty

### Fairness & Ethical Modeling Concepts (Strong Emphasis in HR)

- Adverse Impact Analysis (Four-Fifths Rule) — statistical test for whether a selection process disproportionately screens out a protected group
- Fairness Metrics (Demographic Parity, Equalized Odds) — quantitative checks to ensure predictive models don't produce discriminatory outcomes across protected groups
- Explainability (SHAP, LIME) — essential for HR stakeholders and legal teams to understand and defend why a model produced a given output

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## 08. Resources

### Public Datasets for Practice

- Kaggle: "IBM HR Analytics Employee Attrition & Performance" — the most widely used practice dataset for attrition modeling
- Kaggle: "HR Analytics: Job Change of Data Scientists" — for candidate/employee movement prediction practice
- Kaggle: "Employee Attrition Dataset" (various) — additional practice datasets for classification modeling
- US Bureau of Labor Statistics (BLS) datasets — for broader labor market context and benchmarking practice

### Recommended Courses

- People Analytics courses (Coursera — University of Pennsylvania/Wharton, University of Michigan)
- HR Analytics Specialization (Coursera/edX — various providers)
- Compensation and Benefits courses (Coursera/LinkedIn Learning)
- SHRM People Analytics certification programs

### Books

- "The Power of People: Learn How Successful Organizations Use Workforce Analytics To Improve Business Performance" — Nigel Guenole, Jonathan Ferrar, Sheri Feinzig
- "Data-Driven HR" — Bernard Marr
- "Work Rules!" — Laszlo Bock (former Google People Operations lead, foundational perspective on people analytics culture)
- "People Analytics in the Era of Big Data" — Jeffrey M. Stanton, Robert D. Rogelberg

### Communities & Blogs

- People Analytics World / RWA (The Josh Bersin Company) — industry research and events
- Kaggle competition discussions on HR/attrition datasets
- Towards Data Science (Medium) — search "people analytics", "HR data science" articles
- LinkedIn groups: "People Analytics", "HR Data Science Professionals"

## Standards & Reference Material

- ISO 30414 — Human Capital Reporting standard, increasingly referenced for workforce disclosures
- SHRM (Society for Human Resource Management) research and benchmarking reports
- EEOC Uniform Guidelines on Employee Selection Procedures (US) — foundational reference for adverse impact analysis

## GitHub Repositories for Reference

Search GitHub for: employee-attrition-prediction, hr-analytics-dashboard, pay-equity-analysis for open-source implementation examples.

*This completes the Human Resources (HR Analytics) domain notes.*